Hermetiq

Buildbarn on Kubernetes

Sharing what we've learned operating Buildbarn



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Apache-2.0 charts github.com/Hermetiq/hermetiq-k8s

You already know this work



Observability
Relating build failures to infrastructure signals

Capacity
Scaling from queue demand; checking readiness

Data
Choosing storage and sizing the index

Identity
Connecting OIDC, Bazel credentials and policy

Edge
Keeping long-running gRPC streams healthy

Security
Making privileged exceptions explicit

Infrastructure
Adapting recipes to your cluster and cloud

Changes
Validating configuration before rollout

Day 2
Replacing nodes and diagnosing incidents

Developer experience
Checking the stack with real workloads

Compare our choices with yours. Borrow what helps. Tell us what we missed.

github.com/Hermetiq/hermetiq-k8s

Buildbarn on Kubernetes should be a paved road, not a research project.

You've already learned a lot of these lessons.
We're sharing ours as charts, checks and recipes.

Borrow what helps your deployment.

github.com/Hermetiq/hermetiq-k8s



Our packaging choices, built on upstream

Upstream reference

Kustomize base
Frontend and scheduler
Sharded storage on PVs
Ubuntu worker example
Shared Jsonnet configuration

Unmodified ghcr.io/buildbarn images.
No Buildbarn fork.

31 objects in one example render

4 **workloads**
Frontend, scheduler, browser, storage

4 **ConfigMaps**
Core, worker, JWKS and key-sync config

5 **auth objects**
Key-sync jobs, ServiceAccount and RBAC

5 **routing objects**
gRPC / HTTP routes and edge policies

6 **observability objects**
Five scrapes and one rule set

7 **availability objects**
Five Services and two disruption budgets

Example configuration: frontend KEDA off, RBE web route on. Worker pools are separate RbeWorker resources.

hermetic by nature

Our struggle with Buildbarn configuration

1 The struggle

We were figuring out Buildbarn's Jsonnet configuration.

2 We asked Claude

It kept suggesting configuration structures that didn't exist.

3 The extra work

Understand Buildbarn.
Then work out which AI answers were real.

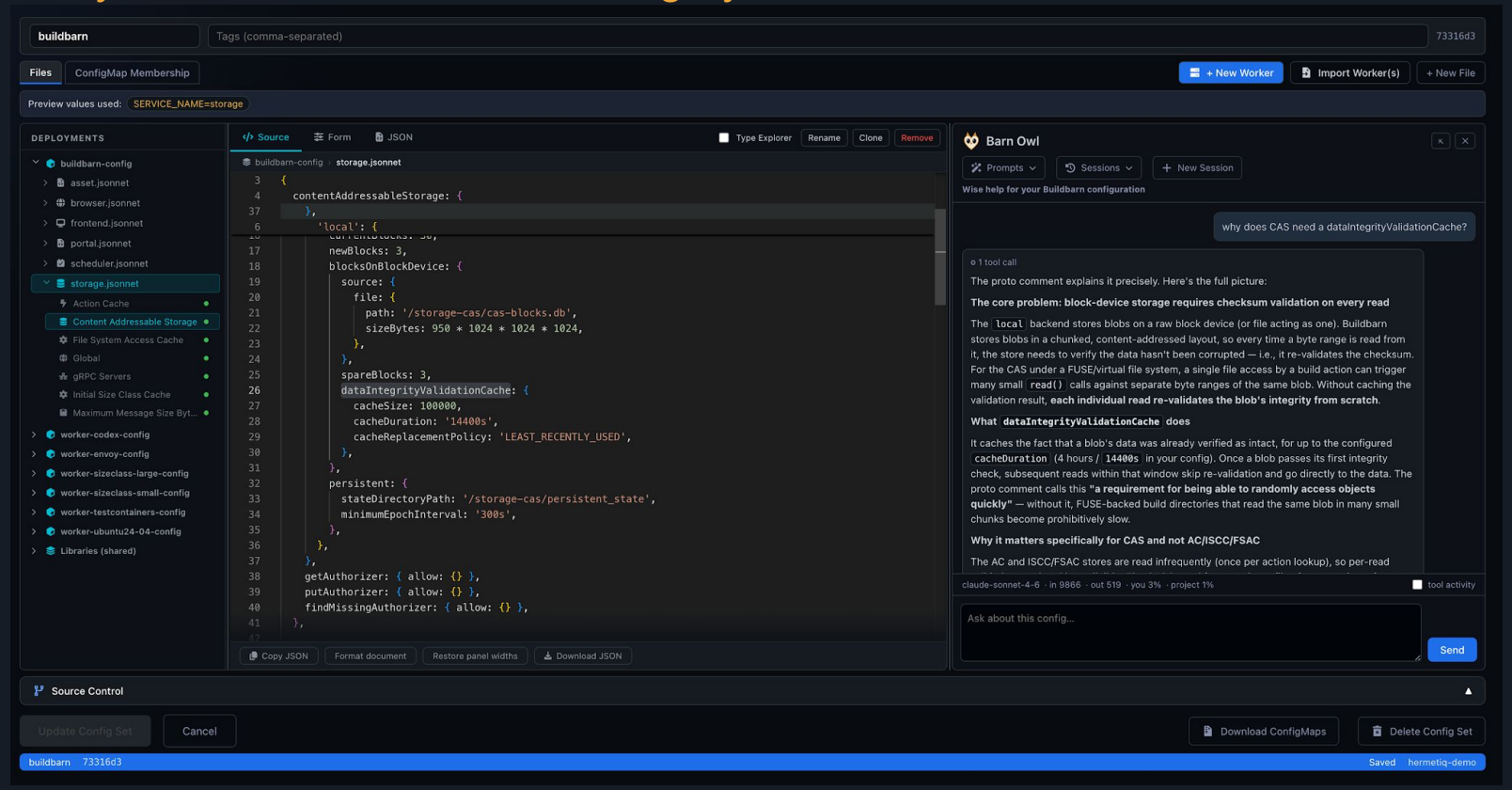
We needed help grounded in real Buildbarn configuration.

github.com/Hermetiq/hermetiq-k8s

owl-driven development
a.k.a. ODD

So we built Barn Owl

Why does CAS need a dataIntegrityValidationCache?



Buildbarn context

Configuration
The settings you're working with

Schema
Configuration and storage rules

Telemetry
Queues, workers and storage

Bazel data
Build events, profiles and actions

Barn Owl proposes. A human commits. Helm applies.

Storage tradeoffs you can compare



Decision	Options	Operational consequence
Persistence	PVC / emptyDir / hostPath	Remote disk or node-local lifetime
Store backend	Filesystem / raw block	volumeMode: Block for raw devices
Key-location map	In memory / file-backed	Restart behavior and metadata I/O
Stores	CAS + AC; optional ISCC / FSAC	Artifacts, results, size classes, prefetch

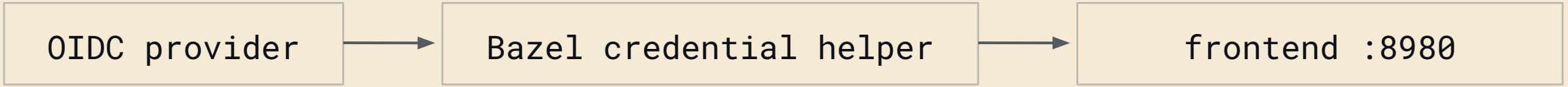
Three GKE block-device recipes

Persistent Disk / Hyperdisk Local NVMe via LVM TopoLVM CSI

hostPath raw block requires an explicit privileged configuration. TopoLVM avoids that storage-container privilege.

remote execution.
local peace
of mind.

Our SSO recipe, with a Dex example



Starter values: reads open; JWT required for CAS put, AC put and Execute.

JWKS syncs into a ConfigMap. Frontends reload keys every 300 s.
Browser SSO via oauth2-proxy. Internal mTLS needs compatible config and mounted Secrets.

Dex

Example OIDC provider
for a test installation

```
requestTimeout: 0s
maxStreamDuration: 0s
8192 pending / 8192 parallel
```

Gateway API, HTTPRoute-only, Contour, Ingress or none

Dex starter included. Connect your identity provider; configure Bazel credentials and routing for your environment.

github.com/Hermetiq/hermetiq-k8s

Why worker pools get an operator



Buildbarn core

A fixed set of services
Changes at release cadence
Consistency checked at render time

Helm

Worker pools

An open set of toolchains and sizes
Replicas move under KEDA / HPA
Config and health reconcile continuously

RbeWorker + controller

One owner for replicas. A Helm upgrade should not reset an autoscaled pool.

If the operator stops: pool reconciliation pauses. Existing serving components and KEDA / HPA keep running.

One pool, one RbeWorker

```yaml
apiVersion: bb.hermetiq.com/v1
kind: RbeWorker
metadata: {name: worker-ubuntu22-04}
spec:
  config:
    generated:
      commonConfigMapName: buildbarn-worker-config
      concurrency: 11
      platformProperties:
      - name: container-image
        value: >-
          docker://ghcr.io/catthehacker/ubuntu:act-22.04
    rolloutOnChange: true
  autoscaling:
    enabled: true
    minReplicas: 1
    maxReplicas: 20
    prometheus: {threshold: "11"}
```

Separate images

Pinned separately. The action's platform selects the matching pool.

Pod placement

Node selectors, tolerations, runtime class, worker / runner limits.

Reconciled objects

Deployment, ConfigMap and KEDA ScaledObject. Status exposes owned names.

Excerpt, not a complete manifest. config.files accepts custom Jsonnet; CEL requires exactly one configuration mode.

1000 workers.
1 owl.
0 pages.

Desired workers aren't ready workers

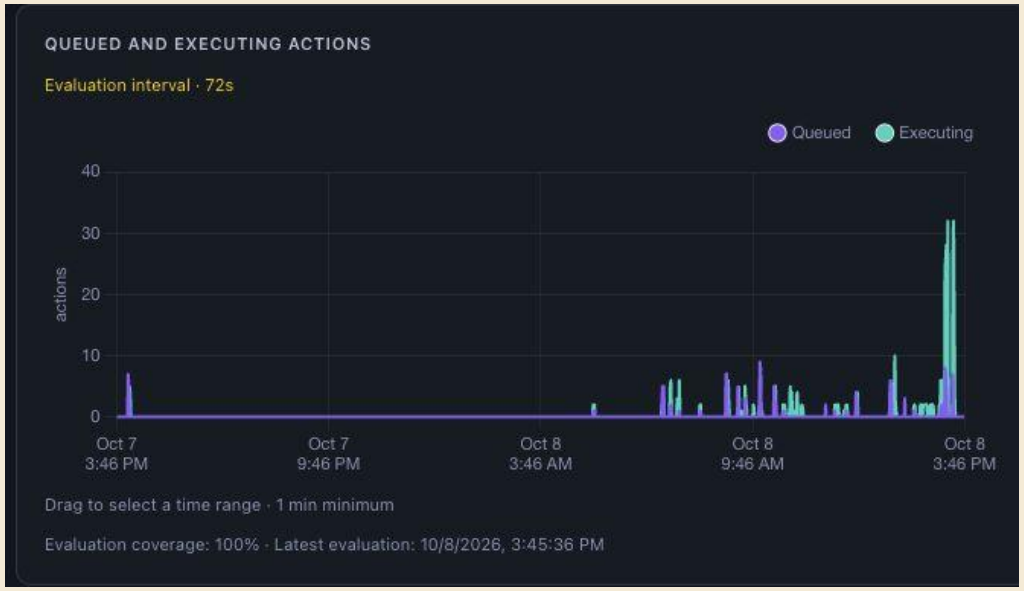
Use both views: worker capacity and scheduler demand.

Hermetiq RBE Infrastructure · Buildbarn namespace · Captured 8 Oct 2026

Worker Pools: desired / ready



Scheduler: queued / executing



Requested capacity
≠ ready capacity

Watch queue depth and ready capacity together.

Worker snapshot + scheduler history; not a matched scale-up timeline.

github.com/Hermetiq/hermetiq-k8s



Our defaults—and their exceptions

Security boundaries

Pod contexts

Non-root, seccomp and dropped capabilities on core services.

Unauthenticated internal ports

Optional NetworkPolicies for storage and scheduler.

Worker credentials

No service-account token by default. Narrow init capabilities.

Explicit exceptions

FUSE, Docker-in-Docker and hostPath block storage need privilege.

Questions we can answer

Scrapes and recording rules

VMPodScrape + VMRule; configure annotation discovery for Prometheus.

Cache health

Eviction age, dropped puts, exhausted gets and retention alerts.

Dashboards in the bundle

Buildbarn dashboards ship with the hermetiq chart.

Incident diagnostics

Optional OTel traces; pprof and active spans on diagnostics ports.

Working toward CIS benchmark alignment. Privileged workloads require explicit exceptions; no compliance claim is implied.

✓ Core ✓ Workers ○ Guardrails ○ Owl

github.com/Hermetiq/hermetiq-k8s

still migrating
to bzlmod
(send help)

Take what helps your deployment

Real workloads

Abseil C++
Envoy
OpenAI Codex
Testcontainers: Redis and Vespa
Drake

We use these charts daily for
managed Buildbarn and our own CI.

Ways to use this work

1. Compare defaults with your deployment
2. Borrow a storage or identity recipe
3. Evaluate worker-pool management
4. Try the examples in a test cluster
5. Share gaps and improvements

Kubernetes 1.32+. GKE-oriented examples.
Adapt storage, identity and routing for your cloud.

Useful ideas can travel independently of the charts. Buildbarn and worker pools run without the Hermetiq application.

✓ Core ✓ Workers ✓ Guardrails ○ Owl

github.com/Hermetiq/hermetiq-k8s

Investigate the Buildbarn you already run



Connect Bazel build data to the Buildbarn serving it.

Build events + profiles + action data + infrastructure telemetry

Understand the build

Investigate failures and slow actions

Explore cache behavior and critical paths

Track flaky tests and trends over time

Understand the infrastructure

Connect queue delays to worker capacity

Inspect storage and execution signals

Bring that context to Barn Owl

Keep your Buildbarn. Add Hermetiq’s build intelligence.

Available views depend on connected telemetry. The Buildbarn charts work independently of the Hermetiq application.

A build engineer agent that knows your stack.

Buildbarn telemetry

Understand queues, workers and storage alongside the builds they serve.

Buildbarn configuration

Explain settings and propose configuration changes your team can review.

Bazel build data

Use build events, profiles and action data to investigate what is slowing you down.

One agent. Connected context. More efficient build infrastructure.

Barn Owl proposes. A human commits. Helm applies.

Bring us one Buildbarn question. See Barn Owl in action.

Bring us a problem from your stack

Keep your Buildbarn. Add Hermetiq.

Bring us a slow build, an unexplained cache miss,
or a worker queue you can't explain.

Prefer managed? Try Hermetiq Cloud.

github.com/Hermetiq/hermetiq-k8s

Learn about the chart. Meet us at the booth.
Grab some stickers!



PEEL RESPONSIBLY

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RBE Infrastructure

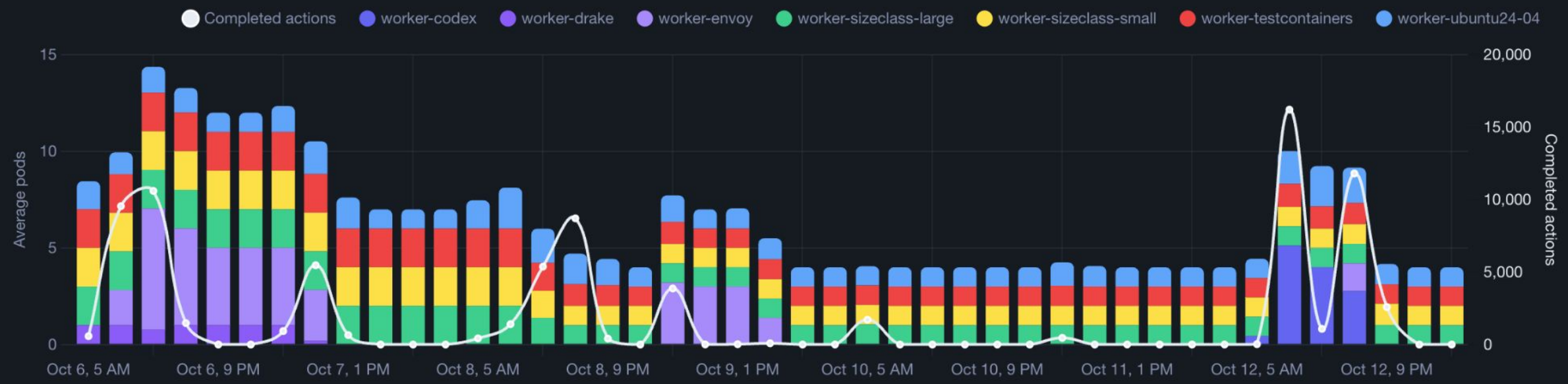
Scheduler health, worker capacity, configuration and cost · buildbarn

Ask Barn Owl

6h | 12h | 24h | 7d | Custom Range | America/Los_Angeles | 1m | Refresh | Open Grafana ↗

Overview | Scheduler | Worker Pools | Worker Nodes | Cost

WORKER PODS BY POOL



Click a bar or drag to zoom to a time range

Average replicas in each 4 h interval. A gap means no replica samples, not zero pods.

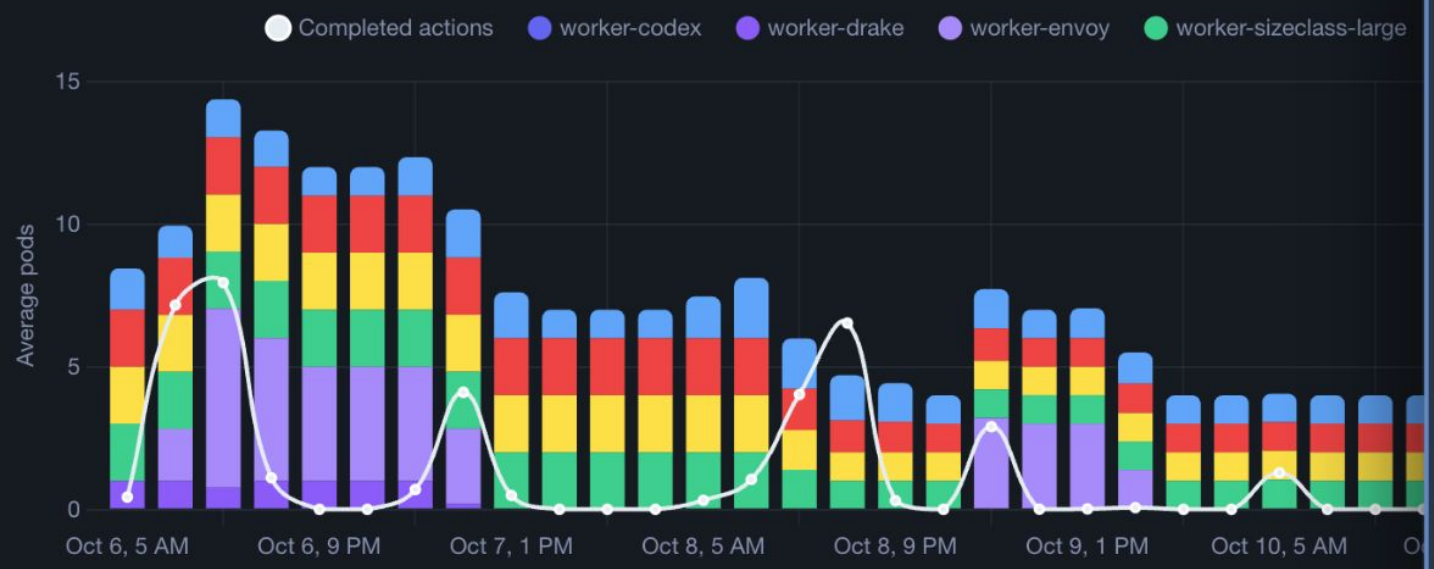
RBE Infrastructure

Scheduler health, worker capacity, configuration and cost · buildbarn

6h 12h 24h 7d Custom Range America/Los_Angeles 1m Refresh

Overview Scheduler Worker Pools 1 Worker Nodes Cost

WORKER PODS BY POOL



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Barn Owl

Ask about queue health, autoscaling, or worker performance

infra-bound. Graph restructuring would not have fixed a queue backed by zero replicas.

How you'd actually get what you asked for

Run the build with `--build_event_json_file` and `--profile`, then join on action digest: BEP gives you `cached: true/false` per action and the critical path; the profile gives you the queue/fetch/exec/upload split. That join is where "time wasted on misses" becomes a real number. It's a Bazel-side workflow, outside what I can execute.

If you have an **invocation ID** from this project, I can scope every metric above to that execution window — far more diagnostic than a 24h aggregate, though still server-side only.

Want me to check `worker-envoy`'s autoscaling bounds and whether its platform string actually matches that queue? That's answerable with what I have, and it's where the 240s tail is coming from.

▶ Prompt details

Was this helpful?

claude-opus-5 · in 32141 · out 2062 · you 0% · project 100% tool activity

Ask Barn Owl about this fleet or selected pool...

Send